
BECK'S TRIAD
SPEED KILLS
PMI
DISS ECTION
SPEED KILLS TOUR
Y NO DISSENT
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2.7 - Pericardial Effusion, Cardiac Tamponade, Constrictive Pericarditis SOAPs

1) Guitar case full of water:
Pericardial effusion = extra fluid within the pericardial space

2) Happy guy:
Most patients with pericardial effusions **won't have symptoms** specific to the effusion **(Chief Complaint)**

3) SPEED KILLS:
Acute pericardial effusions will produce symptoms at a smaller volume **(HPI)**

4) Red-lined guitar case:
Acute pericarditis is a common cause of pericardial effusion **(HPI)**

5) Cancer crab-shaped guitar:
Malignancy (especially metastatic cancer) can cause pericardial effusion without pericarditis **(HPI)**

6) Rusty scalpel-shaped bass:
Chest or cardiac **surgery** can cause pericardial effusion **(HPI)**

7) Arm cast:
Chest wall trauma or type A aortic dissection can cause hemopericardium **(HPI)**

8) DISS-ECTION shirt:
Chest wall trauma or **type A aortic dissection** can cause hemopericardium **(HPI)**

9) Heart-shaped earmuffs on roadie:
Muffled heart sounds may be present in large pericardial effusions **(Physical Exam)**

10) Unable to feel PMI bass:
Inability to palpate the PMI is seen in large pericardial effusions **(Physical Exam)**

11) Alternating sound mixer:
Electrical alternans = cyclic beat-to-beat shift in QRS axis associated with swinging of the heart → seen in large pericardial effusions **(ECG)**

12) Volume <5:
Low QRS voltage = amplitude ≤ 5mm in each of the limb leads → seen in large pericardial effusions **(ECG)**

13) Skull & X-bones t-shirt:
In large pericardial effusions, **CXR** may show cardiomegaly & classic "water bottle" silhouette **(CXR)**

14) Red water bottle:
In large pericardial effusions, CXR may show cardiomegaly & classic **"water bottle" silhouette (CXR)**

15) Heart-shaped megaphone:
Echocardiogram = test of choice for diagnosis & monitoring of pericardial effusion **(Echo)**

16) Hand squeezing heart:
Cardiac tamponade = buildup of intrapericardial pressure that impairs cardiac function

17) SPEED KILLS tour:
Cardiac tamponade is much more likely to occur with **ACUTE pericardial effusions**

18) Guy blowing puff of air:
Cardiac tamponade most commonly presents with **dyspnea** and chest discomfort **(Chief Complaint)**

19) Guy being hit in the chest:
Cardiac tamponade most commonly presents with dyspnea and **chest discomfort (Chief Complaint)**

20) Fainting singer with BP cuff:
Hypotension with narrow pulse pressure is often seen in cardiac tamponade **(Vital Signs)**

21) Blue double-neck guitar:
Elevated JVP with **distended neck veins** is frequently seen in cardiac tamponade **(Physical Exam)**

22) Heart-shaped earmuffs on metal drummer:
Muffled heart sounds are sometimes heard in cardiac tamponade due to presence of pericardial effusion **(Physical Exam)**

23) Beck's Triad:
Beck's Triad = hypotension, distended neck veins, muffled heart sounds → only seen in a minority of cases of cardiac tamponade **(Physical Exam)**

24) Heart clock:
Tachycardia is seen in almost all patients with cardiac tamponade **(Vital Signs)**

25) Y NO DISSENT:
JVP waveform will demonstrate **absent y descent** (i.e., impaired ventricular filling) in cardiac tamponade **(Physical Exam)**

26) Pulsus Paradoxus t-shirt:
Pulsus paradoxus = decrease in systolic pressure >10 mmHg during INSPIRATION **(Physical Exam)**

27) BP cuff in shape of 10:
Pulsus paradoxus = **decrease in systolic pressure >10 mmHg** during INSPIRATION **(Physical Exam)**

28) Bouncer holding breath:
Pulsus paradoxus = decrease in systolic pressure >10 mmHg **during INSPIRATION (Physical Exam)**

29) Lightning shield-shaped guitar:
Cardiac tamponade can cause **obstructive shock** → look for hypotension w/ narrow pulse pressure **(Vital Signs)**

30) Lace leading to heart:
In cardiac tamponade, **cardiac catheterization** will show equalization of diastolic pressures in all chambers **(Cardiac Cath)**

31) Heart with equal sign:
In cardiac tamponade, cardiac catheterization will show **equalization of diastolic pressures** in all chambers **(Cath)**

32) Cardiac Tamponade DDx clipboard:
For further workup of Cardiac Tamponade, see Cardiogenic & Obstructive Shock

33) Stabbed in heart with syringe:
Pericardiocentesis is the treatment of choice in most patients with cardiac tamponade **(1)**

34) Stepping into scalpel-shaped guitar case:
Surgical fluid removal should be performed in patients with cardiac tamponade due to hemopericardium **(2)**

35) Constrictive apron:
Constrictive pericarditis = scarring and loss of normal elasticity of the pericardial sac

36) Floppy heart balloon:
Constrictive pericarditis commonly presents with symptoms of **right-sided heart failure** (e.g., peripheral edema, abdominal distention 2/2 ascites) **(Chief Complaint)**

37) Wide floppy pant legs:
Constrictive pericarditis commonly presents with symptoms of right-sided heart failure (e.g., **peripheral edema**, abdominal distention 2/2 ascites) **(Chief Complaint)**

38) Bulging belly:
Constrictive pericarditis commonly presents with symptoms of right-sided heart failure (e.g., dyspnea on exertion, peripheral edema, **abdominal distention 2/2 ascites**) **(Chief Complaint)**

39) Waiter blowing puff of air:
Constrictive pericarditis impairs cardiac output in response to exertion → patients commonly present with **exertional dyspnea (Chief Complaint)**

40) Red-lined guitar case leaning against stage:
Constrictive pericarditis most commonly occurs as a chronic sequela of **acute pericarditis (PMHx)**

41) Radiation symbol:
Constrictive pericarditis often occurs **post-radiation therapy** to the chest (i.e., for Hodgkin disease or breast cancer) **(PMHx)**

42) Cavitary cacti:
TB is a common cause of constrictive pericarditis in endemic areas **(PMHx)**

43) Dirty scalpel on heart-shaped plate:
Prior cardiac surgery is a known cause of constrictive pericarditis **(Past Surgical Hx)**

44) Blue JVP jug:
Patients with constrictive pericarditis often have **elevated JVP (Physical Exam)**

45) Falling Y-shaped martini glass:
In constrictive pericarditis, the JVP waveform will demonstrate **prominent y descent** (i.e., rapid ventricular filling) **(Physical Exam)**

46) Sniffing Kussmaul cookies in blue jar:
Kussmaul's sign (i.e., lack of decrease or increase in JVP with inspiration) is commonly seen in constrictive pericarditis **(Physical Exam)**

47) Waiter tackled in the liver:
Hepatojugular reflux (i.e., sustained elevation in JVP during compression of the liver) is seen in cardiac causes of volume overload (e.g., constrictive pericarditis), but NOT in hepatic causes (e.g., cirrhosis) **(Physical Exam)**

48) Metal fan knocking on table:
A**pericardial knock** (i.e., extra heart sound during early diastole, occuring before than an S3) can sometimes be heard in patients with constrictive pericarditis **(Cardiac Auscultation)**

49) White-lined guitar case:
In some patients with constrictive pericarditis, CXR may show **pericardial calcification (CXR)**

50) Heart-shaped megaphone yelling at guy with guitar case:
Echocardiogram = diagnostic test of choice for constrictive pericarditis → may show pericardial thickening and impaired diastolic filling **(Echo)**

51) Impaired filling of glasses:
In constrictive pericarditis, look for **abrupt impairment of ventricular filling** during diastole **(Echo)**

52) Constrictive Pericarditis DDx clipboard:
For further workup of Constrictive Pericarditis, see Chronic HF

53) Surgery scalpel:
Pericardiectomy = definitive treatment for patients with chronic constrictive pericarditis **(1)**